EXT4 - format and project guide

FILESYSTEM BUILD GUIDE

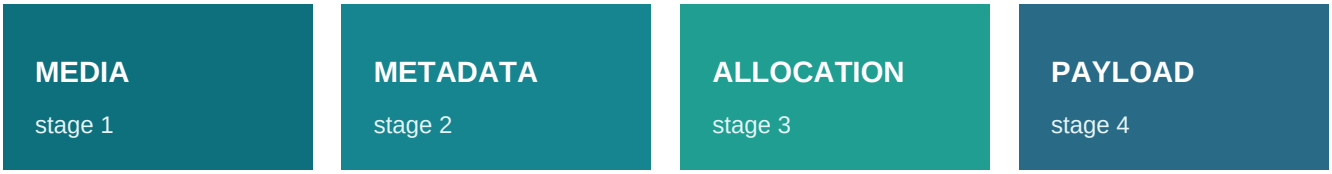
Portable Debian package and locally compiled native installer.

WHAT IS IT?

Linux ext-family filesystem with extents, journaling and compatible on-disk feature flags.

CONCEPTUAL ON-DISK STRUCTURE

Format-specific structures vary; this is not a sector map.



SOURCE AND QUALIFICATION

Filesystem-EXT4 has an independent Infiltrator-Projects source repository. The Debian package records the source revision and payload checksums. Driver capabilities and media compatibility are described in the source qualification reports.

TARGET INSTALLATION LAYOUT

/System/Filesystems/EXT4.so

/System/Resources/Documentation/Filesystem-EXT4/

EXT4.pdf

The installation page gives the exact packaged file paths.

Engineering source and qualification reports belong in GitHub.

Build and installation

Filesystem-EXT4

Release 1.0.3

Installed files

```
/System/Filesystems/EXT4.so

/System/Resources/Documentation/Filesystem-EXT4/EXT4.pdf

/System/Kernel/Filesystems/ext4.ko
```

Debian package

Install the prebuilt files with dpkg -i followed by the Debian filename.

Manual RUN

Run bash followed by the native installer filename.

The installer compiles the shared object and generates this manual locally.

It also compiles the Linux adapter against the running kernel headers.

Use --kernel-build DIRECTORY to select another prepared kernel build.

The manual installer replaces all three files.

InfiltratorOS installation

The OS installer builds a missing or older shared object.

It verifies the manual independently and repairs it from its matching release.

The kernel module is stored for manual use under its single filename.

The package metadata records the source revision and SHA-256 checksums.